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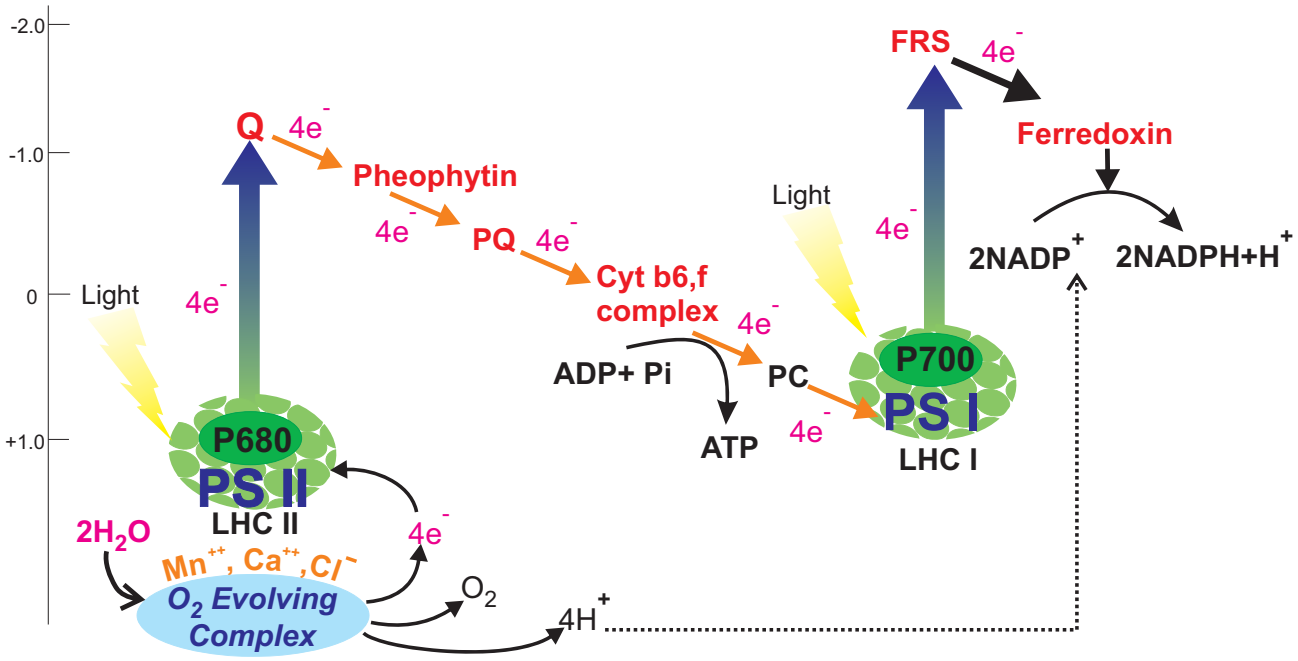


Figure 13.16: Non-Cyclic Photophosphorylation

water molecules both PS I and PS II are activated (Table 13.3). Non-cyclic electron transport PS I and PS II both are involved co-operatively to transport electrons from water to $NADP^+$ (Figure 13.6). In oxygenic species non-cyclic electron transport takes place in three stages.

i. Electron transport from water to P680:

Splitting of water molecule produce electrons, protons and oxygen. Electrons lost by the PS II (P680) are replaced by electrons from splitting of water molecule.

ii. Electron transport from P680 to P700:

Electron flow starts from P680 through a series of electron carrier molecules like pheophytin, plastoquinone (PQ), cytochrome b_6-f complex, plastocyanin (PC) and finally reaches P700 (PS I).

iii. Electron transport from P700 to $NADP^+$

PS I(P700) is excited now and the electrons pass to high energy level. When electron travels downhill through ferredoxin, $NADP^+$ is reduced to NADPH + H^+.

13.11.3 Bio energetics of light reaction

- To release one electron from pigment system it requires two quanta of light.
- One quantum is used for transport of electron from water to PS I.
- Second quantum is used for transport of electron from PS I to $NADP^+$
- Two electrons are required to generate one NADPH + H^+.
- During Non-Cyclic electron transport two NADPH + H^+ are produced and it requires 4 electrons.
- Transportation of 4 electrons requires 8 quanta of light.

Check your grasp!

Name the products produced from Non-Cyclic photophosphorylation?

Why does PS II require electrons from water?

Can you find the difference in the Pathway of electrons during PS I and PS II?